

Lily Zhang

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RESEARCH INTERESTS

Noncommutative geometry and Poisson geometry. Particular interests include index theory, group quasi-representations, and the geometry of Lie groupoids, Lie algebroids, and foliations.

EDUCATION

Texas A&M University

College Station, TX

Ph.D. in Mathematics

Present

- Advisor: Prof. Guoliang Yu

Washington University in St. Louis

St. Louis, MO

Master of Arts in Mathematics

May 2026

- Advisor: Prof. Xiang Tang
- Thesis: *A Generalization of Molino's Theory to Riemannian Groupoids*

PREPRINTS

- [1] *A Generalization of Molino's Theory to Riemannian Groupoids*, 2026, [arXiv:2609.23341](https://arxiv.org/abs/2609.23341).

PAPERS IN PROGRESS

- [1] *Group quasi-representations and stability*, in preparation.
- [2] *Explicit representatives for K-theory classes of Roe and localization algebras of Euclidean spaces*, in preparation.
- [3] *Normal quotients and basic cohomology of blow-ups of proper Lie groupoids*, in preparation.
- [4] *Linear holonomy and linearization of singular foliations*, in preparation.

HONORS & AWARDS

- College Advancement Fellowship, Texas A&M University *Jan 2026*
- Charles J. Turck Presidential Honor Scholarship *2019–2023*
- Finalist Award, Mathematical Contest in Modeling (MCM) *May 2022*

GRANTS

- JMM Undergraduate Travel Grant *Jan 2023*
- Macalester College Wallace Scholarly Activities (WSA) Program *2022–2023*
- MAA MathFest Student Scholars Travel Grant *Aug 2022*

CONTRIBUTED TALKS & POSTERS

- Index Theory Seminar, Texas A&M University *Sep 2026*
- Analytic K-homology Seminar, Texas A&M University *Summer 2026*
- Poisson 2026, KU Leuven, Belgium *Aug 2026*
- Noncommutative Geometry Seminar, Washington University in St. Louis *May 2026*

- Szegő Seminar, Washington University in St. Louis *Oct 2025*
- Joint Mathematics Meetings (JMM), Boston, MA *Jan 2023*
- MAA MathFest, Philadelphia, PA *Aug 2022*
- SIMIODE EXPO, Virtual *Feb 2022*

CONFERENCES & WORKSHOPS ATTENDED

- Groundwork for Operator Algebras Lecture Series, East Lansing, MI *Jun 2026*
- 46th Annual Great Plains Operator Theory Symposium, Iowa City, IA *May 2026*
- Workshop on Poisson Geometry (Gone Fishing), Montréal, QC *Apr 2026*
- Recent Progress in Topological and Geometric Structures in Low Dimensions, Berkeley, CA *Mar 2026*
- Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Berkeley, CA *Jan 2026*
- AMS 2025 Fall Central Sectional Meeting, Saint Louis University, St. Louis, MO *Oct 2025*

TEACHING EXPERIENCE

Texas A&M University, Department of Mathematics *College Station, TX*

- TA, MATH 150 – Functions, Trigonometry and Linear Systems *Fall 2026*

Washington University in St. Louis, Department of Mathematics *St. Louis, MO*

- Grader, MATH 5015 – Introduction to Fourier Series and Integrals *Fall 2025*
- Grader, MATH 415 – Partial Differential Equations *Fall 2025*
- Grader, MATH 309 – Matrix Algebra *Spring 2025*
- Grader, MATH 429 – Proof-Based Linear Algebra *Fall 2024*

Macalester College, Department of Mathematics, Statistics & Computer Science *St. Paul, MN*

- TA, MATH 377 – Real Analysis *Fall 2021*

Ross Mathematics Program *Hybrid: Columbus, OH & Remote*

- Junior Counselor *Summer 2020*

SERVICE

- Instructor, Texas A&M Math Circle *Present*
- Secretary, Association for Women in Mathematics (WUAWM), Washington University in St. Louis *2025–2026*
- Co-organizer, Szegő Seminar, Washington University in St. Louis *2025–2026*
- Speaker, Washington University Math Circle, St. Louis, MO *2024–2025*
- Mentor, MSCS Mentoring Program, Macalester College, St. Paul, MN *2022–2023*